

Supermarket Sales Analysis (Detailed Documentation)

1. Key Performance Indicators (KPIs)

- **Total Revenue:** \$322,967
- **Total Profit:** \$15,379
- **Total Quantity Sold:** 5,510 units
- **Total Invoices:** 1,000
- **Total 5% Markup:** \$15,379

➔ These KPIs provide a high-level overview of business performance, showing strong revenue generation with relatively modest profit margins.

2. Revenue Insights

- **By City:**
 - Naypyitaw: \$110,569 (highest)
 - Yangon: \$106,201
 - Mandalay: \$106,198 ➔ Revenue is evenly distributed across the three cities, with Naypyitaw slightly ahead.
- **By Product Line:**
 - Food & Beverages: highest revenue contributor
 - Sports & Travel: second highest
 - Electronic Accessories & Fashion Accessories: mid-range
 - Home & Lifestyle, Health & Beauty: lowest revenue ➔ Food & Beverages dominate sales, while Health & Beauty lags behind.
- **By Gender:**
 - Female: 51.98%
 - Male: 48.02% ➔ Female customers contribute slightly more revenue overall.

3. Customer & Payment Distribution

- **Customer Type:**
 - Members: 51%
 - Normal: 49% ➔ Membership programs are effective, with members forming the majority of customers.

- **Payment Methods:**

- Cash: 34.74%
- Ewallet: 34.06%
- Credit Card: 31.20% → Cash and ewallet are nearly equal in usage, while credit card payments are slightly less common.

4. Profitability & Trends

- **Profit by City:**

- Naypyitaw: 5,265.2
- Yangon: 5,057.2 → Naypyitaw leads in profitability, closely followed by Yangon.

- **Daily Trends (Revenue & Profit):**

- Revenue fluctuates between \$0–\$50K daily.
- Profit varies up to 300 units, showing peaks and troughs. → Daily performance is volatile, highlighting the need for consistent sales strategies.

- **Revenue & Profit by Branch:**

- Branches A, B, and C all generate ~100K revenue.
- Profit levels remain low compared to revenue. → Strong sales across branches, but profit margins are thin.

5. Price–Quantity Relationship

- **Scatter Plot Analysis:**

- Higher average unit costs correlate with lower average quantities sold.
- Categories like Electronic Accessories and Fashion Accessories show this negative relationship clearly. → Pricing strategy directly impacts sales volume — lower prices drive higher quantities.

6. Conclusion

The Supermarket Sales Analysis dashboard reveals:

- Balanced customer base (members vs normal, male vs female).
- Food & Beverages as the top revenue driver.
- Cash and ewallet dominating payment methods.

- City-level parity in revenue, with Naypyitaw slightly leading in both revenue and profit.
- Daily fluctuations in revenue and profit, requiring close monitoring.
- Clear price–quantity tradeoffs, suggesting opportunities for optimized pricing strategies.

This analysis provides actionable insights for **marketing, inventory management, and pricing decisions** to improve profitability while sustaining strong revenue growth.